

PUMP CAVITATION INVESTIGATION GUIDE

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Asset Class: Centrifugal Pump

Purpose: Cavitation Detection, Root Cause Analysis, Reliability Assessment and Corrective Action Planning

1. Purpose

This guide provides a structured methodology for identifying, diagnosing, investigating and eliminating cavitation in refinery centrifugal pumps.

The document serves as a reference for:

- Operators
 - Maintenance Engineers
 - Reliability Engineers
 - Process Engineers
 - Industrial AI Copilot
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2. What is Cavitation?

Cavitation occurs when liquid pressure falls below the vapor pressure of the process fluid.

When this happens:

- Vapor bubbles form
- Bubbles travel through the pump
- Bubbles collapse violently
- Shockwaves are generated

These shockwaves can damage:

- Impeller surfaces
 - Wear rings
 - Bearings
 - Mechanical seals
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3. Why Cavitation Matters

Cavitation is one of the most destructive failure mechanisms in centrifugal pumps.

Potential consequences include:

- Flow reduction
 - Efficiency loss
 - Excessive vibration
 - Impeller erosion
 - Seal failure
 - Bearing failure
 - Production loss
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4. Cavitation Physics

Step 1

Suction pressure decreases.

Step 2

Local fluid pressure drops below vapor pressure.

Step 3

Vapor bubbles form.

Step 4

Fluid enters higher pressure zones.

Step 5

Bubbles collapse violently.

Step 6

Micro-shockwaves impact metal surfaces.

Step 7

Surface erosion develops.

Step 8

Pump reliability deteriorates.

5. Telemetry Signature of Cavitation

Primary Indicators

Telemetry:

suction_pressure

flow_rate

vibration

Typical Pattern

Suction Pressure:

4.5 → 3.8 → 2.9 → 1.8 bar

Flow Rate:

320 → 290 → 250 → 190 m³/hr

Vibration:

2 → 4 → 6 → 8 mm/s

Secondary Indicators

Telemetry:

bearing_temp

motor_current

Observed Effects

Bearing Temperature:

Gradual increase

Motor Current:

Moderate increase

6. Cavitation Progression Model

Stage 1 — Early Cavitation

Characteristics:

Occasional vapor formation.

Minimal damage.

Telemetry

Suction Pressure:

3.5 – 5 bar

Flow Rate:

Above 280 m³/hr

Vibration:

2 – 4 mm/s

Failure Probability:

10 – 30%

RUL:

180+ Days

Recommended Action

Monitor.

Review suction system.

Stage 2 — Developing Cavitation

Characteristics

Frequent bubble formation.

Initial erosion.

Telemetry

Suction Pressure:

2.5 – 3.5 bar

Flow Rate:

240 – 280 m³/hr

Vibration:

4 – 6 mm/s

Failure Probability:

30 – 60%

RUL:

90 – 180 Days

Recommended Action

Investigate suction conditions.

Stage 3 — Severe Cavitation

Characteristics

Significant impeller damage begins.

Telemetry

Suction Pressure:

1.8 – 2.5 bar

Flow Rate:

190 – 240 m³/hr

Vibration:

6 – 8 mm/s

Failure Probability:

60 – 90%

RUL:

30 – 90 Days

Recommended Action

Immediate reliability review.

Stage 4 — Critical Cavitation

Characteristics

Rapid equipment degradation.

Potential secondary failures.

Telemetry

Suction Pressure:

<1.8 bar

Flow Rate:

<190 m³/hr

Vibration:

8 mm/s

Failure Probability:

90%

RUL:

<30 Days

Recommended Action

Immediate corrective action.

7. Common Root Causes

Cause 1

Blocked Suction Strainer

Symptoms:

Low suction pressure

Reduced flow

Increasing vibration

Cause 2

Low Tank Level

Symptoms:

Intermittent cavitation

Flow instability

Cause 3

Partially Closed Suction Valve

Symptoms:

Pressure reduction

Flow reduction

Cause 4

Undersized Suction Line

Symptoms:

Persistent cavitation

Reduced efficiency

Cause 5

Excessive Fluid Temperature

Symptoms:

Increased vapor formation

Unstable operation

8. NPSH Investigation

Definition

NPSH

Net Positive Suction Head

Represents available pressure above fluid vapor pressure.

Requirement

Available NPSH

Must exceed

Required NPSH

If

Available NPSH < Required NPSH

Result:

High cavitation risk

9. Inspection Procedure

Step 1

Review Telemetry

Analyze:

- Suction Pressure
 - Flow Rate
 - Vibration
 - Bearing Temperature
-

Step 2

Inspect Suction Strainer

Check:

- Debris
 - Fouling
 - Restriction
-

Step 3

Verify Valve Positions

Check:

- Isolation valves
 - Control valves
-

Step 4

Inspect Tank Conditions

Verify:

- Fluid level
 - Fluid temperature
-

Step 5

Calculate NPSH Margin

Determine:

Available NPSH

Required NPSH

10. Cavitation Damage Assessment

Inspect:

- Impeller leading edges
 - Wear rings
 - Casing surfaces
-

Typical Damage Patterns

Pitting

Surface erosion

Material removal

11. AI Copilot Diagnostic Logic

If:

Suction Pressure ↓

Flow Rate ↓

Vibration ↑

Diagnosis:

Cavitation

Confidence:

High

If:

Suction Pressure ↓

Flow Rate ↓

Vibration ↑

Bearing Temperature ↑

Diagnosis:

Advanced Cavitation

Potential Secondary Bearing Damage

Confidence:

Very High

12. AI Copilot Severity Classification

Low Risk

Failure Probability <30%

Action:

Monitor

Medium Risk

Failure Probability 30–60%

Action:

Inspect suction system

High Risk

Failure Probability 60–90%

Action:

Reliability engineer review

Critical Risk

Failure Probability >90%

RUL <30 Days

Action:

Immediate corrective action

13. Corrective Action Matrix

Root Cause	Corrective Action
Blocked Strainer	Clean Strainer
Low Tank Level	Restore Inventory
Closed Valve	Open Valve
High Temperature	Reduce Temperature
NPSH Deficiency	Increase Available NPSH

14. Example Investigation Case

Observed Telemetry

Suction Pressure:

1.7 bar

Flow Rate:

185 m³/hr

Vibration:

8.3 mm/s

Bearing Temperature:

82°C

Failure Probability:

92%

RUL:

18 Days

Diagnosis

Severe Cavitation

Probable Cause

Suction Restriction

Recommended Actions

1. Inspect Strainer
 2. Verify Valve Position
 3. Verify NPSH Margin
 4. Inspect Impeller
 5. Schedule Maintenance
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15. Lessons Learned

Cavitation is frequently misdiagnosed as bearing failure because both conditions increase vibration.

Always evaluate:

- Suction Pressure
- Flow Rate
- Vibration

together before concluding the failure mechanism.

Proper diagnosis significantly improves maintenance effectiveness and prevents unnecessary component replacement.

16. Summary

Cavitation is characterized by:

- Low Suction Pressure
- Reduced Flow Rate
- Elevated Vibration

The Industrial AI Copilot shall use these indicators along with Failure Probability and RUL to identify cavitation, determine severity and recommend corrective actions before significant equipment damage occurs.